II

The path of institutional change

I now turn to two fundamental questions of societal, political, and economic change. First, what determines the divergent patterns of evolution of societies, polities, or economies over time? And how do we account for the survival of economies with persistently poor performance over long periods of time?

If we look back far enough in history, divergence appears to be very simple to explain. Bands and tribes confronted different problems with different resource endowments, different human capabilities, and in different climates. Out of these emerged different solutions to the common problems of survival, including different languages, customs, traditions, and taboos. There is no reason to believe that solutions should be similar, although there is reason to believe that they would tend to converge over time as the cost of information fell. However, after ten thousand years of civilization, despite the immense decline in information costs and despite the implications of neoclassical international trade models that would suggest convergence, there is enormous contrast between economies.

Which brings me to the second issue. What accounts for the survival of societies and economies that are characterized by persistent poor performance? Since Charles Darwin, evolutionary theory has had a powerful influence upon our understanding of social survival, and it has been embedded in the literature of economics since the publication of Armen Alchian's 1950 article. The implications of the theory are that over time inefficient institutions are weeded out, efficient ones survive, and thus there is a gradual evolution of more efficient forms of economic, political, and social organization.

I have used the term efficient in this study to indicate a condition where the existing set of constraints will produce economic growth. Specifically, institutions that enable the parties in the exchange to capture more of the gains from trade will grow relative to those that fail to realize this potential. Either emigration to the more successful economies or emulation of

the institutions of those economies would result. Again going back to the Coase theorem: in a world of zero transaction costs, the efficient solution that produced the highest aggregate income would prevail. But because transaction costs are not zero, we could anticipate differential performance reflecting different degrees of success of institutional frameworks in reducing transaction (and transformation) costs. But why would the relatively inefficient economies persist? What prevents them from adopting the institutions of the more efficient economies?

If institutions existed in the zero transaction cost framework, then history would not matter; a change in relative prices or preferences would induce an immediate restructuring of institutions to adjust efficiently, as described in Chapter 2 on the competitive model. But if the process by which we arrive at today's institutions is relevant and constrains future choices, then not only does history matter but persistent poor performance and long-run divergent patterns of development stem from a common source.

I

As a first approximation to dealing with these issues, I turn to an interesting body of economic literature that has focused primarily on the evolution of technology, but has made analogies to a broader range of questions, including, although mostly implicitly, institutional change. The article that first called the attention of economic historians to the issue of path dependence is Paul David's "Clio and the Economics of QWERTY" (1985). In this article, David attempts to explain how the peculiar organization of letters on the typewriter keyboard became standardized and fixed and to explain what accidental set of happenings appears to have caused this result to persist, even in the face of more efficient alternatives. Technological anomalies of this kind are not hard to find. The persistence of narrow-gauge rails, the success of alternating current over direct current, and the survival of the gas engine over steam engine motor cars have all been used to illustrate the peculiar fact that incremental changes in technology, once begun on a particular track, may lead one technological solution to win out over another, even when, ultimately, this technological path may be less efficient than the abandoned alternative would have been.

The argument that small historical events can lead one technology to win out over another was developed first by W. Brian Arthur.[1] I elaborate

[1]For a brief survey of Arthur's arguments and a summary of a substantial amount of his work, see his "Self-Reinforcing Mechanisms in Economics," in a volume called *The Economy as An Evolving Complex System* (1988).

the argument along the lines he set down. Let us examine, side by side, two competing technologies that both yield increasing returns. Agents apply learning by doing to these separate technologies, improving the efficiency of each in a manner analogous to the way organizations evolve (see Chapter 9). Each agent adapts more efficient ways of solving problems and of utilizing new technologies and equipment, and yet we may not be able to predict which technology will turn out to be the most efficient one. Because the rate of increasing returns may not remain constant for both, however, they may not grow at the same rate. Moreover, subsequent breakthroughs in one technology, unknown to the players originally, may result in monopolistic domination of the other because increasing returns imply a single winner over time. Or, simply, some small event may give one technology an advantage over the other. Hence, one technology will win out and maintain a monopolistic position, even though its successful innovations may turn out, downstream, to be inferior (or a blind alley) compared to the abandoned alternative technology. Arthur has in mind four self-reinforcing mechanisms: (1) large setup or fixed costs, which give the advantage of falling unit costs as output increases; (2) learning effects, which improve products or lower their costs as their prevalence increases; (3) coordination effects, which confer advantages to cooperation with other economic agents taking similar action; and (4) adaptive expectations, where increased prevalence on the market enhances beliefs of further prevalence.[2]

The consequence of these self-reinforcing mechanisms is, in Arthur's terms, characterized by four properties: (1) multiple equilibria – a number of solutions are possible and the outcome is indeterminate; (2) possible inefficiencies – a technology that is inherently better than another loses out because of bad luck in gaining adherence; (3) lock-in – once reached, a solution is difficult to exit from; (4) path dependence – the consequence of small events and chance circumstances can determine solutions that, once they prevail, lead one to a particular path.

Can one extend this argument of technological change to institutional change? To review its assumptions: Arthur deals with competitive markets in which agents respond to maximizing opportunities; he is analyzing competing technologies, both of which are subject to increasing returns. In fact (although I am not aware that Arthur makes this distinction), the competition is only indirectly between technologies. Directly it is between organizations embodying the competing technologies. The distinction is important because the outcome may reflect differing organizational abilities (tacit knowledge of the entrepreneurs) as much as specific aspects of the competing technologies. Indeed, ultimately Arthur is

[2]Arthur (1988), p. 10.

dealing with decision making in organizations, as is the institutional model of this study.

II

There are two forces shaping the path of institutional change: increasing returns and imperfect markets characterized by significant transaction costs. Although Arthur's technological story is coextensive with the first, neither he nor David explicitly deals with the second. I shall deal with them in turn.

In a world in which there are no increasing returns to institutions and markets are competitive, institutions do not matter. If, as discussed in Chapter 2, the actors initially have incorrect models and act upon them, they either will be eliminated or efficient information feedback will induce them to modify their models.

But, with increasing returns, institutions matter. Indeed, all four of Arthur's self-reinforcing mechanisms apply, although with somewhat different characteristics. There are large initial setup costs when the institutions are created de novo as was the U.S. Constitution in 1787. There are significant learning effects for organizations that arise in consequence of the opportunity set provided by the institutional framework (as elaborated in Chapter 9). The resultant organizations will evolve to take advantage of the opportunities defined by that framework, but as in the case of technology, there is no implication that the skills acquired will result in increased social efficiency. There will be coordination effects directly via contracts with other organizations and indirectly by induced investment through the polity in complementary activities. Even more important, the formal rules will result in the creation of a variety of informal constraints that modify the formal rules and extend them to a variety of specific applications. Adaptive expectations occur because increased prevalence of contracting based on a specific institution will reduce uncertainties about the permanence of that rule. In short, the interdependent web of an institutional matrix produces massive increasing returns.

With increasing returns, institutions matter and shape the long-run path of economies, but as long as the consequent markets are competitive or even roughly approximate the zero-transaction-cost model, the long-run path is an efficient one as that term has been used here. Given reasonably noncontroversial assumptions about preferences, neither divergent paths nor persistently poor performance would prevail. But if the markets are incomplete, the information feedback is fragmentary at best, and transaction costs are significant, then the subjective models of actors modified both by very imperfect feedback and by ideology will shape the path. Then, not only can both divergent paths and persistently poor

performance prevail, the historically derived perceptions of the actors shape the choices that they make. In a dynamic world characterized by institutional increasing returns, the imperfect and fumbling efforts of the actors reflect the difficulties of deciphering a complex environment with the available mental constructs – ideas, theories, and ideologies.

We return to the institutional evolution occurring in medieval and early modern Western Europe briefly described in Chapter 10. The radical decline in population in the fourteenth century altered the bargaining strength of peasants vis-à-vis lords and led to incremental alterations over time in the implicit contracts between them. The margins at which alterations occurred can only be understood in terms of the historically derived costs of transacting and the historically derived *models* that both parties possessed about their worlds. The transaction costs were embodied in the customs of the manor, which had evolved over time in defining the relationship of lord to serf. The historically derived model that each possessed of his world included a status relationship of inequality characterized by a master-servant status; neither party would have even envisioned a change that would have eliminated that inequality. The incremental changes are only intelligible in terms of these historical relationships. If institutions were not subject to increasing returns and subjective perceptions were always corrected to true models, then presumably the actors would immediately have recontracted to a far more efficient joint solution. In fact, because there were increasing returns to the institutional framework, the process was incremental and, as described earlier, consisted of a slow evolution of formal and informal constraints and enforcement changes. And because, in this particular instance, competitive political forces and very slowly changing mental constructs of the status of both parties combined to produce more efficient outcomes (both in agriculture and in commerce), we tell it as a success story entitled *The Rise of the Western World*.

But it is still an exceptional story in economic history (see Chapter 13). Throughout most of history the experience of the agents and the ideologies of the actors do not combine to lead to efficient outcomes. Before systematically examining the sources of persistently inefficient paths, I shall attempt to make the process of path dependence clearer with several illustrations.

III

The evolution of the common law, a form of institutional change, is helpful in understanding overall institutional change. Common law is precedent based – it provides continuity and essential predictability that are critical to reducing uncertainty among contracting parties. Past deci-

sions become embedded in the structure of law, which changes marginally as new cases arise involving new, or at least in terms of past cases unforeseen, issues; when decided these become, in turn, a part of the legal framework. The judicial decisions reflect the subjective processing of information in the context of the historical construction of the legal framework. Now if, in fact, the common law is efficient, as a number of modern law and economics scholars have asserted, it would be because the competitive process does indeed lead the judicial actors to correct models. But if the judicial decision makers operate on the basis of incomplete information and their subjective and ideologically conditioned views of how the world ought to be, then no such assertion should be made.[3] However we account for the judicial process, the institutional framework is being continuously but incrementally modified by the purposive activities of organizations bringing cases before the courts.

The Northwest Ordinance, a specific legislative enactment, illustrates the historically derived continuity implied by path dependence as well as the downstream consequences of increasing returns. The act itself was a law of fundamental importance to the development of the polity and economy of the United States. It was passed in 1787 by the Continental Congress at the very time that the Constitutional Convention was meeting in Philadelphia. The ordinance was the third act to deal with a whole range of issues concerned with the governance and settlement of the vast area of land in the West and provided a framework by which the territories would be integrated into the new nation. It will be useful to describe the ordinance, where the rules came from, how they were incorporated, and how they relate to the issues of path dependence.

The ordinance is quite simple and brief. It provided for rules of inheritance and fee-simple ownership of land and it set up the basic structure of the territorial governments and provided for the mechanisms by which territories gradually became self-governing. Additionally, it made provisions for when a territory could be admitted as a state. Then there were a series of articles of compact, in effect a bill of rights for the territories (i.e., provisions for religious freedom, the writ of habeas corpus, trial by jury, bailment, enforcement of contract, and compensation for property). There were additional provisions about good faith to the Indians, free navigation on the Mississippi and the St. Lawrence, public debt, land

[3]In "Imperfect Decisions and the Law: On the Evolution of Legal Precedent and Rules," Heiner (1986) makes the point forcefully in his analysis of the evolution of common law that as agents increasingly must interpret less familiar "non-local" (to use Heiner's term) information, they process it imperfectly. Hence, legal precedent establishes relatively simple standards that a judge can follow. Such a conclusion is in sharp contrast to the *efficient* consequences of common law that are characteristic of much of the law and economics research.

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disposal, the number of states that could be divided up within the Northwest Territory, and finally a provision prohibiting slavery (though the return of runaway slaves was provided for) in the territories.

It is easy to trace the source of most of the provisions. The subjective models of the authors of the ordinance can be directly traced to the historical evolution of English and colonial thought (Hughes, 1987). The specific provisions had become a part of the rules of political units of the colonies during the previous 150 years. These included inheritance laws, fee-simple ownership of land, and many of the provisions of the Bill of Rights. Some, however, although precedent-based, had become controversial because legislators foresaw that the organizations (in this case states) that they represented would be affected by them – for example, provisions about the size of new states and the conditions for their admittance. The precedence was derived from the original provisions of charters and from the Articles of Confederation, but controversies arose because the conditions for admitting territories into states would critically influence the relative power and bargaining strength of existing states. One of the rules, the prohibition of slavery, appears to have been the result of vote trading between the authors of the Northwest Ordinance and the writers of the Constitution; slavery was prohibited in the former bill in return for counting slaves as three-fifths of a person in the Constitution, which increased the representation of Southern slave states in Congress (a major issue of the period).

The Northwest Ordinance provided the basic framework dictating the pattern of expansion of the United States over the next century. Although its provisions were at times modified by new issues and controversies, it provided a clear, path-dependent pattern of institutional evolution. The increasing-returns characteristics stemmed from the fact that the structure of property rights, inheritance laws, and political decision rules in the territories was derived from the act and in turn spawned organizations and (political and economic) entrepreneurs who induced marginal alterations in the act downstream. Indeed, the very success of the act was reflected in the growing influence of new Western territories and states and successful efforts by their representatives to modify land policy in their interests (North and Rutten, 1987). Therefore, U.S. land history is only understandable as a story of incremental institutional change involving interplay between the institutional framework and the consequent organizations.

If, however, the foregoing story sounds like an inevitable, foreordained account, it should not. At every step along the way there were choices – political and economic – that provided real alternatives. Path dependence is a way to narrow conceptually the choice set and link decision making through time. It is not a story of inevitability in which the past neatly

predicts the future. In the story briefly recounted above, the pieces of the bill were in part derived from the colonial charters, but the final bill was significantly altered as a result of (1) conflicts among the states over entry conditions of the territories (which would determine the subsequent bargaining position of existing states), (2) the North/South issues over slavery, and (3) the coincidental Constitutional Convention in Philadelphia.

We can now integrate the path-dependent character of the incremental change in institutions with the persistence of patterns of long-run growth or decline. Once a development path is set on a particular course, the network externalities, the learning process of organizations, and the historically derived subjective modeling of the issues reinforce the course. In the case of economic growth, an adaptively efficient path, as described in Chapter 9, allows for a maximum of choices under uncertainty, for the pursuit of various trial methods of undertaking activities, and for an efficient feedback mechanism to identify choices that are relatively inefficient and to eliminate them. Note that the Northwest Ordinance not only provided adaptively efficient economic development – by fee-simple ownership of land and a clear system of inheritance that in turn made the transferability of land possible at low transaction costs – it also provided an efficient system of government, which allowed for the political transaction costs of integrating the territories into the national government to be low. Indeed, it is not too much to say that despite the inefficiencies of some specific subsequent land acts in the nineteenth century, the basic provisions of the Northwest Ordinance provided for relatively efficient solutions to these problems with the easy transfer of land, so that no matter how inappropriately we may have devised land distribution schemes later on, their costs were minimized to a substantial degree by the basic provisions of the Northwest Ordinance.

But so, too, can unproductive paths persist. The increasing returns characteristic of an initial set of institutions that provide disincentives to productive activity will create organizations and interest groups with a stake in the existing constraints. They will shape the polity in their interests. Such institutions provide incentives that may encourage military domination of the polity and economy, religious fanaticism, or plain, simple redistributive organizations, but they provide few rewards from increases in the stock and dissemination of economically useful knowledge. The subjective mental constructs of the participants will evolve an ideology that not only rationalizes the society's structure but accounts for its poor performance. As a result the economy will evolve policies that reinforce the existing incentives and organizations. Thus, both the writings of the Economic Commission for Latin America (ECLA) and dependency theory explain the poor performance of Latin American economies on the basis of the international terms of trade with industrial countries

and other conditions external to those economies. Such an explanation not only rationalizes the structure of Latin American economies, but also contains policy implications that would reinforce the existing institutional framework.

Because all economies have institutional frameworks that create both productive and unproductive opportunities for organizations, the history of any economy will reflect some mixed results. Recall that the immediate instruments of institutional change are political or economic entrepreneurs who attempt to maximize at those margins that appear to offer the most profitable (short-run) alternatives. Whether the most promising alternative is investing in piracy, constructing an oil cartel, or developing a more high-powered chip for computers, it is the existing constraints and changes in incentives at the margin that determine opportunities. But note that the agent – the entrepreneur – not only is constrained in alternatives by the existing institutions, but has imperfect knowledge with respect to accomplishing his or her objective. Therefore, even if – a *big* if – the objective happened to be consistent with increasing productivity, there is no guarantee that the goal would be realized, and unexpected consequences could lead to radically different results (a technological breakthrough that made property rights more insecure or increased the payoffs to terrorism, for example). In effect, short-run efforts at profit maximizing may result in the pursuit of persistently inefficient activities (given the institutional constraints) and, even if they pursue productive activities, may have unexpected consequences. (This can, of course, work in the other direction too; pirates might eventually find that settlement and trade turned out to be more profitable, as the Vikings did.)

However, it would be a mistake to think that successful paths get reversed by small events or errors and vice versa. Recall the increasing-returns nature of the institutional matrix made up of a complex of interdependent rules and informal constraints that in total determine economic performance; individual, specific changes in formal or informal constraints certainly may change history, but for the most part do not reverse its direction. The brief account of United States land policy makes clear that although specific acts were inefficient, the overall institutional framework (comprising not only the Northwest Ordinance, but the two preceding ordinances, the complementary provisions embodied in the United States Constitution, and the equally complementary informal constraints that had evolved) reduced their inefficient consequences.

Path dependence means that history matters. We cannot understand today's choices (and define them in the modeling of economic performance) without tracing the incremental evolution of institutions. But we are just beginning the serious task of exploring the implications of path dependence.

IV

Why does a fundamental change in relative prices affect two societies differently? The answer should now be clear. In each society the change will result in adaptations at the margin, and the margins affected will be those where the immediate issues require solution and the solution will be determined by the relative bargaining power of the participants – that is, the organizations that have evolved in the specific overall institutional context. But note that it will be a marginal adjustment, built upon the preceding institutional arrangements. Because the bargaining power of groups in one society will clearly differ from that in another, the marginal adjustments in each society will typically be different as well. Moreover, with different past histories and incomplete feedback on the consequences, the actors will have different subjective models and therefore make different policy choices. Marginal adjustment in such cases does not lead to convergence.

What happens when a common set of rules is imposed on two different societies? I can illustrate from an historical example. The U.S. Constitution was adopted (with modifications) by many Latin American countries in the nineteenth century, and many of the property rights laws of successful Western countries have been adopted by Third World countries. The results, however, are not similar to those in either the United States or other successful Western countries. Although the rules are the same, the enforcement mechanisms, the way enforcement occurs, the norms of behavior, and the subjective models of the actors are not. Hence, both the real incentive structures and the perceived consequences of policies will differ as well. Thus, a common set of fundamental changes in relative prices or the common imposition of a set of rules will lead to widely divergent outcomes in societies with different institutional arrangements.

V

The focus of this chapter has been on gradual institutional change occurring through continuous marginal adjustments. The emphasis on this type of change is deliberate. It is the dominant way by which societies and economies have evolved. But as briefly discussed in the preceding chapter, discontinuous institutional change by conquest or revolution is also important. Such institutional discontinuities only reinforce my argument, however, because the tenacious survival of institutional constraints in the face of radical alterations in the formal rules of the game is the best evidence of the increasing-returns characteristics of an institutional framework. For example, take the revolutions that swept North and South America and created independence from Britain and Spain in the

eighteenth and early nineteenth centuries. The evolution of North America and of Latin America differed radically right from the beginning, reflecting the imposition of the institutional patterns from the mother country upon the colonies and the radically divergent ideological constructs that shape the perceptions of the actors.

In the case of North America, the English colonies were formed in the very century when the struggle between Parliament and the Crown was coming to a head. Religious diversity as well as political diversity in the mother country were paralleled in the colonies and were reflected in the ideas and models that came to be eloquently articulated in the eighteenth century. There was substantial diversity in the political structure of crown, proprietary, and charter colonies, but the general development in the direction of local political control and the growth of assemblies was clear and unambiguous. Similarly, the navigation acts placed the colonies within the framework of overall British imperial policy. But within that broad framework, the colonists were free to develop their own economy. Indeed, sometimes the colonists themselves imposed more restrictions on property rights than did the mother country.

The French and Indian War (1756 to 1763) is a familiar breaking point in U.S. history. British efforts to impose a very modest tax on colonial subjects, as well as to curb westward migration, produced a violent reaction. The subjective perception of many colonists was that the British navigation acts threatened the prosperity of the colonies. In fact, the burden of the navigation acts was negligible and it is reasonable to presume that had the colonies remained a part of Britain, as Canada did, they would have prospered. But the perception of the colonists was different, and their acting on that perception led – via steps taken by individuals and organizations – to the Revolutionary War, the Declaration of Independence, the Articles of Confederation, the Northwest Ordinance, and the Constitution, a sequence of institutional expressions that formed a consistent evolutionary institutional pattern. Yet although the revolution created the United States, postrevolutionary history is only intelligible in terms of the continuity of informal and, indeed, many formal institutional constraints carried over from before the revolution.

In the case of the Spanish Indies, conquest came at the precise time that the influence of the Castilian Cortes was declining, the conquerors imposed a uniform religion and a uniform bureaucratic administration on an already existing agricultural society (particularly in the highlands of Mexico and Alto Peru, where agricultural societies were well developed), the bureaucracy detailed every aspect of political and economic policy (again much more stringently and effectively applied in the populated and valued regions than in the nomadic and empty areas), and there were recurrent crises over the problems of agency and control of the bureau-

cratic machinery. Although efforts at reversing the centralized bureaucratic policy occurred under the Bourbons and even to some extent led to the liberalization of trade within the empire, the reversal was partial and quickly negated. The control of agents was a persistent problem compounded by the efforts of the Creoles to take over the bureaucracy to pursue their own interests. Although the Wars of Independence turned out to be a struggle for control of the bureaucracy and consequent polity and economy between local colonial control and imperial control, nevertheless the struggle was imbued with the ideological overtones that stemmed from the U.S. and French revolutions. As a consequence, independence brought U.S.-inspired constitutions, but the results were radically different.

In the case of the United States, the Constitution embodied the ongoing heritage of first British and then colonial economic and political policies complemented by a consistent ideological modeling of the issues. In the case of Latin America, an alien set of rules was imposed on a long heritage of centralized bureaucratic controls and accompanying ideological perceptions of the issues. In consequence, Latin American federal schemes and efforts at decentralization did not work after the first few years of independence. The gradual reversion, country by country, to bureaucratic centralized control characterized Latin America in the nineteenth and the twentieth centuries. The persistence of the institutional pattern that had been imposed by Spain and Portugal continued to play a fundamental role in the evolution of Latin American policies and perceptions and to distinguish that continent's history, despite the imposition after independence of a set of rules similar to the British institutional tradition that shaped the path of North America.[4]

VI

Technological change and institutional change are the basic keys to societal and economic evolution and both exhibit the characteristics of path dependence. Can a single model account for both technological and institutional change? They do have much in common. Increasing returns is an essential ingredient to both. The perceptions of the actors play a more central role in institutional than in technological change because ideological beliefs influence the subjective construction of the models that determine choices. Choices are more multifaceted in an institutional context because of the complex interrelationships among formal and informal

[4]For a summary account of the Latin American experience see C. Veliz, *The Centralist Tradition in Latin America* (1980) or W. G. Glade, *The Latin American Economies: A Study of Their Institutional Evolution* (1969).

constraints. In consequence, both lock-in and path dependence appear much more complicated in the case of institutions than in the case of technology. The interplay between the polity and the economy, the many actors who have varying degrees of bargaining strength in influencing institutional change, and the role of cultural inheritance that appears to underlie the persistence of many informal constraints all contribute to this complexity.

I conclude this chapter by emphasizing some implications of this analysis. Long-run economic change is the cumulative consequence of innumerable short-run decisions by political and economic entrepreneurs that both directly and indirectly (via external effects) shape performance. The choices made reflect the entrepreneurs' subjective modeling of the environment. Therefore, the degree to which outcomes are consistent with intentions will reflect the degree to which the entrepreneur's models are *true* models. Because the models reflect ideas, ideologies, and beliefs that are, at best, only partially refined and improved by information feedback on the actual consequences of the enacted policies, the consequences of specific policies are not only uncertain but to a substantial degree unpredictable. Even the most casual inspection of political and economic choices, both throughout history and today, makes clear the wide gap between intentions and outcomes. However, the increasing-returns characteristics of the institutional matrix and the complementary subjective models of the players suggest that although the specific short-run paths are unforeseeable, the overall direction in the long run is both more predictable and more difficult to reverse.

Part III

Economic performance

12

Institutions, economic theory, and economic performance

We cannot see, feel, touch, or even measure institutions; they are constructs of the human mind. But even the most convinced neoclassical economists admit their existence and typically make them parameters (implicitly or explicitly) in their models. Do institutions matter? Do tariffs, regulations, and rules matter? Does government make a difference? Can we explain the radical change in economic well-being when we step across the boundary between the United States and Mexico? What makes markets work or not work? Does honesty in exchange make a difference; does it pay? I hope that the analysis of the previous chapters has provided a convincing framework to shed light on the consequences of institutions.

But I wish to assert a much more fundamental role for institutions in societies; they are the underlying determinant of the long-run performance of economies. If we are ever to construct a dynamic theory of change – something missing in mainstream economics and only very imperfectly dealt with in Marxian theory – it must be built on a model of institutional change. Although some of the pieces of the puzzle are still missing, the outline of the direction to be taken is, I believe, clear.

In the sections that follow I (1) specify what changes must be made in neoclassical theory to incorporate institutional analysis into that theory, (2) outline the implications for the *static* analysis of economic performance, and (3) explore the implications of institutional analysis for the construction of a dynamic theory of long-run economic change.

I

Information processing by the actors as a result of the costliness of transacting underlies the formation of institutions. At issue are both the meaning of rationality and the characteristics of transacting that prevent the actors from achieving the joint maximization result of the zero transaction cost model.

The instrumental rationality postulate of neoclassical theory assumes that the actors possess information necessary to evaluate correctly the alternatives and in consequence make choices that will achieve the desired ends. In fact, such a postulate has implicitly assumed the existence of a particular set of institutions and information. If institutions play a purely passive role so that they do not constrain the choices of the actors and the actors are in possession of the information necessary to make correct choices, then the instrumental rationality postulate is the correct building block. If, on the other hand, the actors are incompletely informed, devise subjective models as guides to choices, and can only very imperfectly correct their models with information feedback, then a procedural rationality postulate (as described in Chapter 3) is the essential building block to theorizing.

The former postulate evolved in the context of the highly developed, efficient markets of the Western world and has served as a useful tool of analysis in such a context. But those markets are characterized by the exceptional condition of low or negligible transaction costs. I know of no way to analyze most markets in the contemporary world and throughout history with such a behavioral postulate. A procedural rationality postulate, on the other hand, not only can account for the incomplete and imperfect markets that characterize much of the present and the past world, but also leads the researcher to the key issues of just what it is that makes markets imperfect. That leads us to the costs of transacting.

The costs of transacting arise because information is costly and asymmetrically held by the parties to exchange and also because any way that the actors develop institutions to structure human interaction results in some degree of imperfection of the markets. In effect, the incentive consequences of institutions provide mixed signals to the participants, so that even in those cases where the institutional framework is conducive to capturing more of the gains from trade as compared to an earlier institutional framework, there will still be incentives to cheat, free ride, and so forth that will contribute to market imperfections. Given the behavioral characteristics of human beings, there is simply no way to devise institutions that solve the complex exchange problems and at the same time are free of some incompatible incentives. As a result, much of the recent literature of industrial organization and political economy has attempted to come to grips with incentive incompatibility in economic and political organization (see Miller, *Managerial Dilemmas: The Political Economy of Hierarchies,* forthcoming). The success stories of economic history describe the institutional innovations that have lowered the costs of transacting and permitted capturing more of the gains from trade and hence permitted the expansion of markets. But such innovations, for the most part, have not created the conditions necessary for the efficient markets of

the neoclassical model. The polity specifies and enforces the property rights of the economic marketplace, and the characteristics of the political market are the essential key to understanding the imperfections of markets.

What would make the political market approximate the zero transaction cost model for efficient economic exchange? The condition is easily stated. Legislation would be enacted which increased aggregate income and in which the gainers compensated losers at a transaction cost that is low enough to make it jointly worthwhile. The informational and institutional conditions necessary to realize such exchange are:

1. The affected parties must have the information and correct model to know that the bill affects them and to know the amount of gains or losses they would incur.
2. The results can be communicated to their agent (the legislator) who will faithfully vote accordingly.
3. Votes will be weighted by the aggregate net gains or losses so that the net result can be ascertained and the losers appropriately compensated.
4. This exchange can be accomplished at a low enough cost of transacting to make it worthwhile.

The institutional structure most favorable to approximating such conditions is a modern democratic society with universal suffrage. Vote trading, log rolling, and the incentive of an incumbent's opponents to bring his or her *deficiencies* before constituents and hence reduce agency problems all contribute to *better* outcomes.

But look at the disincentives built into the system. Rational voter ignorance is not just a buzzword of the public choice literature. Not only could the voter never acquire the information to be even vaguely informed about the myriad bills that affect his or her welfare, but there is no way that the constituent (or even the legislator) could ever possess accurate models to weigh the consequences. Agency theory has provided abundant, if controversial, evidence of the degree to which the legislator acts independently of constituent interests. Whereas the legislator is going to trade votes on the basis of perceived number of votes he or she stands to gain or lose, that is frequently a long way from reflecting net gains or losses to all the constituents. And how often is there an incentive to compensate losers? There is a vast gap between *better* and *efficient* (in the neoclassical meaning of that term) outcomes, as a vast literature in modern political economy will attest. For my purpose, it is necessary to emphasize two essential conditions that loom large. They are that the affected parties have both the information and the correct model to accurately appraise the consequences and that all the affected parties have equal access to the decision-making process. These conditions are not

even approximately met in the most favorable institutional framework in all of history for efficient political decision making.

Because polities make and enforce economic rules, it is not surprising that property rights are seldom efficient (North, 1981). But even when efficient property rights are devised, they will still typically have features that will be very costly to monitor or enforce, reflecting built-in disincentives or at the very least aspects of the exchange that provide temptation to renege, shirk, steal, or cheat. In many cases informal constraints will evolve to mitigate these disincentive consequences. And the modern Western world provides abundant evidence of markets that work and even approximate the neoclassical ideal. But they are exceptional and difficult to come by, and the institutional requirements are stringent.

II

The consequences of institutions for contemporary economic analysis can be summarized as follows:

1. Economic (and political) models are specific to particular constellations of institutional constraints that vary radically both through time and cross sectionally in different economies. The models are institution specific and in many cases highly sensitive to altered institutional constraints. A self-conscious awareness of these constraints is essential both for improved theory construction and for issues of public policy. It is not just how well would the model play in Bangladesh or in the United States during the nineteenth century, but much more immediately, how would it play in another developed country like Japan or even in the United States next year?

Even more important is that the specific institutional constraints dictate the margins at which organizations operate and hence make intelligible the interplay between the rules of the game and the behavior of the actors. If organizations – firms, trade unions, farm groups, political parties, and congressional committees to name a few – devote their efforts to *unproductive activity,* the institutional constraints have provided the incentive structure for such activity. Third World countries are poor because the institutional constraints define a set of payoffs to political/economic activity that do not encourage productive activity. Socialist economies are just beginning to appreciate that the underlying institutional framework is the source of their current poor performance and are attempting to grapple with ways to restructure the institutional framework to redirect incentives that in turn will direct organizations along productivity-increasing paths. And as for the first world, we not only need to appreciate the importance of an overall institutional framework that has been responsible for the growth of the economy, but to be self-conscious about

the consequences of the ongoing marginal changes that are continually occurring– not only on overall performance but also on specific sectors of the economy. We have long been aware that the tax structure, regulations, judicial decisions, and statute laws, to name but a few formal constraints, shape the policies of firms, trade unions, and other organizations and hence determine specific aspects of economic performance; but such awareness has not led to a focusing of economic theory on modeling the political/economic process that produces these results.

2. A self-conscious incorporation of institutions will force social scientists in general, and economists in particular, to question the behavioral models that underlie their disciplines and, in consequence, to explore much more systematically than we have done so far the implications of the costly and imperfect processing of information for the consequent behavior of the actors. Social scientists have incorporated the costliness of information in their models, but have not come to grips with the subjective mental constructs by which individuals process information and arrive at conclusions that shape their choices. There is in economics a (largely) implicit assumption that the actors can correctly identify the reason for their predicaments (i.e., have *true* theories), know the costs and benefits of alternative choices, and know how to act upon them (see, for example, Becker, 1983). Our preoccupation with rational choice and efficient market hypotheses has blinded us to the implications of incomplete information and the complexity of environments and subjective perceptions of the external world that individuals hold. There is nothing the matter with the rational actor paradigm that could not be cured by a healthy awareness of the complexity of human motivation and the problems that arise from information processing. Social scientists would then understand not only why institutions exist, but also how they influence outcomes.

3. Ideas and ideologies matter, and institutions play a major role in determining just how much they matter. Ideas and ideologies shape the subjective mental constructs that individuals use to interpret the world around them and make choices. Moreover, by structuring the interaction of human beings in certain ways, formal institutions affect the price we pay for our actions, and to the degree the formal institutions are deliberately or accidentally structured to lower the price of acting on one's ideas, they provide the freedom to individuals to incorporate their ideas and ideologies into the choices they make. A key consequence of formal institutions is mechanisms, like voting systems in democracies or organizational structures in hierarchies, that enable individuals who are agents to express their own views and to have a very different impact upon outcomes than those implied by the simple interest-group modeling that has characterized so much of economic and public choice theory.

Economic performance

4. The polity and the economy are inextricably interlinked in any understanding of the performance of an economy and therefore we must develop a true political economy discipline. A set of institutional constraints defines the exchange relationships between the two and therefore determines the way a political/economic system works. Not only do polities specify and enforce property rights that shape the basic incentive structure of an economy, in the modern world the share of gross national product going through government and the ubiquitous and ever-changing regulations imposed by it are the most important keys to economic performance. A useful model of the macroaspect or even microaspects of an economy must build the institutional constraints into the model. Modern macroeconomic theory, for example, will never resolve the problems that it confronts unless its practitioners recognize that the decisions made by the political process critically affect the functioning of economies. Although at an ad hoc level we have begun to recognize this, much more integration of politics and economics than has been accomplished so far is needed. This can only be done by a modeling of the political-economic process that incorporates the specific institutions involved and the consequent structure of political and economic exchange.

III

Integrating institutional analysis into *static* neoclassical theory entails modifying the existing body of theory. But devising a model of economic change requires the construction of an entire theoretical framework, because no such model exists. Path dependence is the key to an analytical understanding of long-run economic change. The promise of this approach is that it extends the most constructive building blocks of neoclassical theory – both the scarcity/competition postulate and incentives as the driving force – but modifies that theory by incorporating incomplete information and subjective models of *reality* and the increasing returns characteristic of institutions. The result is an approach that offers the promise of connecting microlevel economic activity with the macrolevel incentives provided by the institutional framework. The source of incremental change is the gains to be obtained by organizations and their entrepreneurs from acquiring skills, knowledge, and information that will enhance their objectives. Path dependence comes from the increasing returns mechanisms that reinforce the direction once on a given path. Alterations in the path come from unanticipated consequences of choices, external effects, and sometimes forces exogenous to the analytical framework. Reversal of paths (from stagnation to growth or vice versa) may come from the above described sources of path alteration, but will typically occur through changes in the polity.

Economic theory and economic performance

I can expand on the sequential characteristics of path dependence by returning to the contrast between the British-North American path and the Spanish-Latin American path, discussed in Chapter 11.

The background

At the beginning of the sixteenth century, England and Spain had evolved very differently. England had developed a relatively centralized feudalism, as a result of the Norman conquest, and had recently established the Tudors with the Battle of Bosworth (1485). Spain, in contrast, had just emerged from seven centuries of Moorish domination of the Iberian Peninsula. It was not a unified country. Although the marriage of Ferdinand and Isabella brought Castile and Aragon together, they continued to maintain separate rules, Cortes, and policies.

However, both England and Spain faced, in common with the rest of the emerging European nation-states, a critical problem: the need to acquire additional revenue to survive in the face of the rising costs of warfare. The king traditionally lived on his own, that is, off the revenue from his estates together with the traditional feudal dues; but these resources were insufficient in the face of the new military technology associated with the effective use of the crossbow, longbow, pike, and gunpowder. This fiscal crisis of the state, first described by Joseph Schumpeter (1954), forced rulers to make bargains with constituents. In both countries, the consequence was the development of some form of representation on the part of constituents (Parliament in England and the Cortes in Spain) in return for revenue. In both countries, the wool trade became a major source of crown revenue. But the consequences of the common relative price change arising from the new military technology were radically different in the two countries. In one, it led to the evolution of a polity and economy that solved the fiscal crisis and went on to dominate the Western world. In the other, in spite of initially more favorable conditions, it led to unresolved fiscal crises, bankruptcies, confiscation of assets, and insecure property rights and to three centuries of relative stagnation.

In England, the tension between ruler and constituent (although the barons at Runnymede might have caviled at that term) surfaced with the Magna Carta in 1215. The fiscal crisis came later with the Hundred Years War. Stubbs describes the consequence as follows: "The admission of the right of parliament to legislate, to enquire into abuses, and to share in the guidance of national policy, was practically purchased by the money granted to Edward I and Edward III." (Stubbs, 1896, p. 599) The subsequent history to 1689 and the final triumph of Parliament is well known.

In Spain, the union of Aragon (comprising approximately Valencia, Aragon, and Catalonia) and Castile joined two very different regions. Aragon had been reconquered from the Arabs in the last half of the

thirteenth century and had become a major commercial empire extending into Sardinia, Sicily, and parts of Greece. The Cortes reflected the interests of merchants and played a significant role in public affairs. In contrast, Castile was continually engaged in warfare, either against the Moors or in internal strife, and although the Cortes existed it was seldom summoned. In the fifteen years after their union, Isabella succeeded in gaining control not only over the unruly warlike barons, but over church policy in Castile as well. Although the role of the Castilian Cortes has, in recent scholarly work, been somewhat upgraded, nevertheless there was a centralized monarchy and bureaucracy in Castile, and it was Castile that defined the institutional evolution of both Spain and Latin America.

The institutional framework

It was not simply centralization or decentralization in the polity that differentiated the two societies. Nevertheless, this feature made a critical difference and was symptomatic of the broad differences in both the polity and the economy. Not only did the Parliament in England provide the beginning of representative government and a reduction in the *rent-seeking* behavior that had characterized the financially hard-pressed Stuart monarchs, but also Parliament's triumph betokened increased security of property rights and a more effective, impartial judicial system.

Spain's polity consisted of a large centralized bureaucracy that "administered the ever-growing body of decrees and juridical directives, which both legitimized the administrative machinery and laid down its course of action" (Glade, 1969, p. 58). Every detail of the economy as well as the polity was structured with the objective of furthering the interests of the crown in the creation of the most powerful empire since Rome. But with the revolt of the Netherlands and the decline in the inflow of New World treasure, the fiscal demands far outstripped revenue, and the result was bankruptcy, increased internal taxation, confiscations, and insecure property rights.

The organizational implications

In England, Parliament created the Bank of England and a fiscal system in which expenditures were tied to tax revenues. The consequent financial revolution not only finally put the government on a sound financial basis, but laid the ground for the development of the private capital market. More secure property rights, the decline of mercantilist restrictions, and the escape of textile firms from urban guild restrictions combined to provide expanding opportunities for firms in domestic and international markets. Both the growing markets and the patent law encouraged the growth of innovative activity. But all this and much more is a familiar story.

In Spain, repeated bankruptcies between 1557 and 1647 were coupled with desperate measures to stave off disaster. War, the church, and administering the complex bureaucratic system provided the major organizational opportunities in Spain and in consequence the military, priesthood, and the judiciary were rewarding occupations. The expulsion of the Moors and Jews, rent ceilings on land and price ceilings on wheat, confiscations of silver remittances to merchants in Seville (who were compensated with relatively worthless bonds called *juros*) were symptomatic of the disincentives to productive activity.

Path dependence

To make the contrasting brief stories convincing illustrations of path dependence would entail an account of the political, economic, and judicial systems of each society as a web of interconnected formal rules and informal constraints that together made up the institutional matrix and led the economies down different paths. It would be necessary to demonstrate the network externalities that limited the actors' choices and prevented them from radically altering the institutional framework. Such an undertaking is far beyond the kinds of existing empirical evidence with which I am familiar. I can only indirectly infer such implications from the evidence.

In a controversial study, *The Origins of English Individualism* (1978), Alan Macfarlane maintains that at least from the thirteenth century the English were different than the traditional picture we possess of peasant societies. The traditional characteristics – patriarchal domination, extended family, low status of women, tight knit and closed peasant villages, self-sufficiency, and the family as the work unit – all were conspicuously absent by the thirteenth century. Instead, Macfarlane paints a picture of a fluid, individualistically oriented set of attitudes involving the structure of the family, the organization of work, and the social relationships of the village community complemented by an array of formal rules dealing with property, inheritance, and the legal status of women. Macfarlane wants to make the point that England was different and that the difference went way back in time, but in doing so he amasses evidence to make clear the complex interdependent network of formal and informal constraints that made for the increasing returns characteristic of path dependence.

The most telling evidence of the increasing returns feature of the Spanish institutional fabric was the inability of the crown and its bureaucracy to alter the direction of the Spanish path in spite of their awareness of the decay and decline overcoming the country. In a century – the seventeenth – Spain declined from the most powerful nation in the Western world since the Roman empire to a second-rate power. The depopulation of the countryside, the stagnation of industry, and the collapse of Seville's trad-

ing system with the New World were paralleled in the political realm by the revolt of Catalonia and Portugal. The proximate cause was recurrent war and a fiscal crisis that led Olivares (1621 to 1640) to pursue the desperate measures that only exacerbated the fundamental problems. Indeed, the policies that were considered feasible in the context of the institutional constraints and perceptions of the actors were price controls, tax increases, and repeated confiscations. As for the perceptions of the actors, Jan De Vries in his study (1976) of Europe in the age of crisis describes the effort to reverse the decline as follows:

> But this was not a society unaware of what was happening. A whole school of economic reformers . . . wrote mountains of tracts pleading for new measures. . . . Indeed, in 1623 a *Junta de Reformacion* recommended to the new King, Philip IV, a series of measures including taxes to encourage earlier marriage (and, hence, population growth), limitations on the number of servants, the establishment of a bank, prohibitions on the import of luxuries, the closing of brothels, and the prohibition of the teaching of Latin in small towns (to reduce the flight from agriculture of peasants who had acquired a smattering of education). But no willpower could be found to follow through on these recommendations. . . . It is said that the only accomplishment of the reform movement was the abolition of the ruff collar, a fashion which had imposed ruinous laundry bills on the aristocracy. (De Vries, 1976, p. 28)

It appears doubtful that instrumental rationality could be applied to the reasoning of the Junta.

Both England and Spain faced fiscal crises in the seventeenth century, but the contrasting paths that they took appear to have reflected deep underlying institutional characteristics of the societies.

The downstream consequences

U.S. economic history has been characterized by a federal political system, checks and balances, and a basic structure of property rights that have encouraged the long-term contracting essential to the creation of capital markets and economic growth. Even one of the most costly civil wars in all of history failed to alter the basic institutional matrix.

Latin American economic history, in contrast, has perpetuated the centralized, bureaucratic traditions carried over from its Spanish/Portuguese heritage. Here is John Coatsworth's characterization of the institutional environment of nineteenth-century Mexico:

> The interventionist and pervasively arbitrary nature of the institutional environment forced every enterprise, urban or rural, to operate in a highly politicized manner, using kinship networks, political influence, and family prestige to gain privileged access to subsidized credit, to aid various strategems for recruiting labor, to collect debts or enforce contracts, to evade taxes or circumvent the

> courts, and to defend or assert titles to lands. Success or failure in the economic arena always depended on the relationship of the producer with political authorities – local officials for arranging matters close at hand and the central government of the colony for sympathetic interpretations of the law and intervention at the local level when conditions required it. Small enterprise, excluded from the system of corporate privilege and political favors, was forced to operate in a permanent state of semiclandestiny, always at the margin of the law, at the mercy of petty officials, never secure from arbitrary acts and never protected against the rights of those more powerful. (Coatsworth, 1978, p. 94)

The divergent paths established by England and Spain in the New World have not converged despite the mediating factors of common ideological influences. In the former, an institutional framework has evolved that permits the complex impersonal exchange necessary to political stability and to capture the potential economic gains of modern technology. In the latter, personalistic relationships are still the key to much of the political and economic exchange. They are a consequence of an evolving institutional framework that produces neither political stability nor consistent realization of the potential of modern technology.